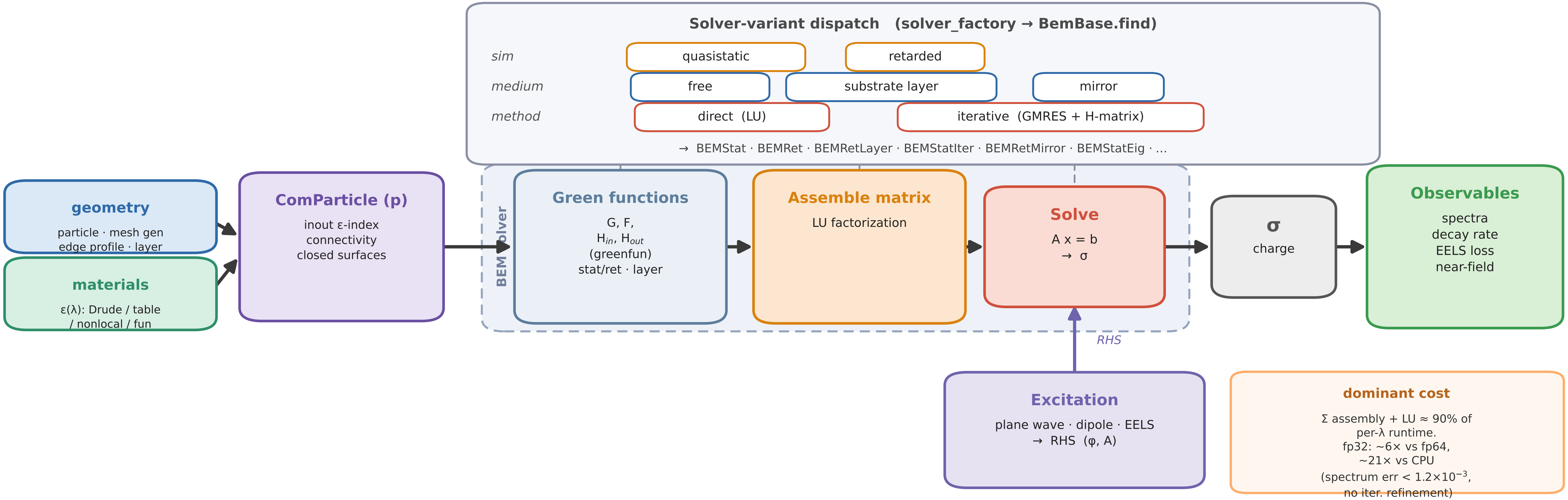
pyMNPBEM — simulation pipeline & compute backend



Compute backend per stage

	Geometry & ε(λ)	ComParticle	Green eval	BEM assemble (Σ, GEMM, LU)	Solve (σ)	Observables / field
CPU (NumPy/SciPy + numba)	✓	✓	✓	✓	✓	✓
GPU fp64 (CuPy, complex128)			△	✓	✓	✓
GPU fp32 (CuPy, complex64)				✓	✓	✓
multi-GPU (cuSolverMg)				✓	✓	

△ Green eval on GPU only for layer (Sommerfeld Bessel/Hankel) & ACA / H-matrix block fill (fp64); dense free-space fill stays on CPU (numba). CPU-resident: mesh & ε eval, refinement quadrature, ACA tree, GMRES residual loop.

mode select: CPU = default | GPU-fp64 = MNPBEM_GPU=1 | GPU-fp32 = + MNPBEM_GPU_LOWPREC=1

multi-GPU = MNPBEM_VRAM_SHARE_GPUS=N (pymnpbem config: compute.gpu_precision = fp32 / fp64)